

Generative Transitions in Cryovolcanic Systems:

A Principia Orthogona Analysis of the Enceladus Plume

Pablo Nogueira Grossi*

June 2026

Preprint — Zenodo 10.5281/zenodo.20779067 — CC BY 4.0

MSC 2020: 37C25, 37G10, 53D10, 57M27

Dedicated to Betül Kaçar, whose resurrection of ancient enzymes showed that the operator chain does not care about the date.

Abstract

The Principia Orthogona framework (Grossi 2026) defines a generative transition as an event in which a trajectory undergoes compression, curvature intensification, loss of injectivity, and stabilisation, governed by the operator sequence $G = U \circ F \circ K \circ C$, with a fifth entropy operator E accumulating irreversible cost. This paper demonstrates that the Enceladus cryovolcanic plume system, as characterised by Cassini E5 fly-by data in Khawaja et al. (*Nature Astronomy* 9, 1662–1671, 2025), constitutes a canonical physical realisation of this sequence.

We derive four mathematical correspondences: (1) the subsurface-to-fissure compression operator C realised by hydrothermal convection channelling a three-dimensional ocean flow into a one-dimensional fracture; (2) the curvature operator K realised by subsurface pressure driving the ice-shell curvature beyond the critical focal radius $\kappa^* \approx (2000 \text{ m})^{-1}$; (3) the fold operator F realised by tiger-stripe fissure ejection, classified as Whitney A_1 (finite branches, Jacobian rank loss by 1); (4) the unfolding operator U realised by plume dispersal and grain orbit insertion into the E ring as a Keplerian attractor Γ . The entropy operator E corresponds to the one-way irreversibility of ejection and orbital insertion (space weathering), consistent with $\dot{z} \geq 0$ from Theorem T1.

*G6 LLC, Newark, New Jersey 07104, USA. ORCID: 0009-0000-6496-2186. Email: g6llc@proton.me

The chemical stratification detected by Khawaja et al. — aromatic and O-bearing species surviving E-ring transit, ether/ethyl compounds absent from the E ring — admits a direct interpretation in terms of Gronwall basin selection: compounds reaching the E-ring attractor satisfy the stability criterion; those absent were ejected by the entropy operator. We identify a new falsifiability condition (F-new) and a new open question (OE-2) testable by a future Enceladus mission.

Contents

1	Introduction	3
2	Mathematical Foundations	3
2.1	Contact Manifolds and the Reeb Field	3
2.2	The Operator Chain $G = U \circ F \circ K \circ C$	4
2.3	Stability Radius $\varepsilon_0 = 1/3$ and the Gronwall Basin	5
2.4	Theorem T1 — Entropy Monotonicity	5
2.5	Cassini Physical Parameters	5
3	Operator Correspondences with Dimensional Estimates	5
3.1	Compression Operator C	5
3.2	Curvature Operator K	6
3.3	Fold Operator F	6
3.4	Unfolding Operator U	7
3.5	Entropy Operator E	7
4	Whitney A_1–A_3 Singularity Classification	7
5	Gronwall Basin Condition and Chemical Survival Record	8
6	Entropy Operator and Theorem T1	9
7	Falsifiability Conditions	9
8	New Open Questions Arising from this Correspondence	10
9	Kalpataru: Life as Generated, Not Originated	10
10	Conclusion	12

1 Introduction

The Principia Orthogona framework identifies a generative transition as a four-phase geometric event: a compression reduces dimensionality, curvature intensifies until a critical threshold is reached, a fold commits the trajectory irreversibly to a new branch, and an unfolding stabilises it at a new attractor. This sequence is not intended as an analogy — it is claimed to be a precise mathematical structure whose instances across different physical domains share the same operator definitions, invariants, and singularity class.

Canonical examples developed in Principia Orthogona Vol. I (Grossi 2026) [2] include cellular autophagy (X_{auto}) and stellar nucleosynthesis via the triple-alpha process (X_{star}). Both are documented in the companion Chapter A [3], where 26 theorems are machine-checked in Lean 4 with zero `sorry`. The Enceladus plume, characterised quantitatively in Khawaja et al. [1] via Cassini Cosmic Dust Analyzer (CDA) mass spectra from the E5 fly-by at 17.7 km s^{-1} , provides a third canonical example with a different physical medium (planetary cryovolcanism), a different spatial scale, and uniquely, an independent chemical record of what the fold event preserved and what it destroyed.

This paper is self-contained. All definitions, operators, the stability radius, and Theorem T1 are stated from first principles in Section 2 so that the paper can be read without consulting any prior volume of the series. Section 3 derives each operator correspondence with dimensional estimates. Section 4 performs the Whitney singularity classification. Section 5 addresses the Gronwall basin and the chemical survival record. Section 6 treats the entropy operator and Theorem T1. Section 7 addresses falsifiability. Sections 8–10 discuss open questions, the generative view of life, and conclusions.

2 Mathematical Foundations

2.1 Contact Manifolds and the Reeb Field

Definition 2.1 (Contact manifold). *A contact manifold is a pair (M, α) where M is a smooth $(2n + 1)$ -dimensional manifold and $\alpha \in \Omega^1(M)$ satisfies*

$$\alpha \wedge (d\alpha)^n \neq 0 \quad \text{everywhere on } M.$$

The contact distribution is the hyperplane field $\xi = \ker(\alpha) \subset TM$.

Definition 2.2 (Reeb vector field). *The Reeb vector field R_α is the unique vector field satisfying*

$$\iota(R_\alpha)\alpha = 1, \quad \iota(R_\alpha)d\alpha = 0.$$

R_α is everywhere transverse to ξ and generates the contact flow.

2.2 The Operator Chain $G = U \circ F \circ K \circ C$

Definition 2.3 (Constraint operator C). *Let (M, α) be a contact manifold with Reeb field R_α . The constraint operator $C : TM \rightarrow \xi$ is*

$$Cv = v - \alpha(v) R_\alpha \quad \text{for } v \in TM.$$

Definition 2.4 (Kinetic / threshold operator K). *Let g be a Riemannian metric on ξ . The kinetic operator $K : \xi \rightarrow \mathbb{R}^+$ is*

$$K(v) = \frac{1}{2} g(v, v), \quad v \in \xi_x.$$

The critical threshold K^ is the smallest value such that $K(v) > K^*$ forces curvature intensification beyond the focal radius.*

Definition 2.5 (Fold operator F — Whitney A_1). *Let $f : M \rightarrow N$ be a smooth map between manifolds of equal dimension. $F = f$ is a Whitney A_1 singularity at $p \in M$ if:*

- (1) $\text{rank}(df_p) = \dim M - 1$ (corank 1);
- (2) $\ker(df_p)$ is transverse to the fold locus $\Sigma^1(f) = \{q : \text{rank } df_q < \dim M\}$;
- (3) the second-order term of f restricted to $\ker(df_p)$ is non-zero.

In local coordinates, the Whitney A_1 normal form is

$$(x_1, x_2, \dots, x_{2n+1}) \mapsto (x_1^2, x_2, \dots, x_{2n+1}).$$

F selects the stable branch $x_1 > 0$ and maps dynamics through the singularity, losing one degree of freedom at the fold locus.

Definition 2.6 (Unfolding operator U). *The unfolding operator $U : N \rightarrow M$ maps the folded output into the Keplerian attractor $\Gamma \subset M$, defined by*

$$\Gamma = \{x \in M : K(v|_x) \leq K^*, V(x) = 0\},$$

where V is the Lyapunov function of Definition 2.8.

Definition 2.7 (Entropy operator E). *The entropy operator E is the vector field X_E on M satisfying*

$$\iota(X_E)d\alpha = dS,$$

where $S : M \rightarrow \mathbb{R}^+$ is the thermodynamic entropy density. E generates the irreversible component of the contact flow.

2.3 Stability Radius $\varepsilon_0 = 1/3$ and the Gronwall Basin

Definition 2.8 (Lyapunov function and stability radius). *Let $r : M \rightarrow \mathbb{R}^+$ be the normalised distance from Γ , with $r|_\Gamma = 1$. Define*

$$V(x) = \frac{1}{2}(r(x) - 1)^2.$$

The Gronwall basin is $\mathcal{B}(\varepsilon_0) = \{x : |r(x) - 1| < \varepsilon_0\}$. The stability radius $\varepsilon_0 = 1/3$ is the maximal ε for which the Gronwall inequality $|\int_0^t f(x(s)) ds| \leq \varepsilon \cdot t$ holds for all orbits $x(t)$ of G starting in $\mathcal{B}(\varepsilon)$.

Remark 2.1 (Chemical interpretation of ε_0). In the Enceladus application (§5), compounds with photodissociation bond energy $D \geq D_{\min}$ survive E-ring transit and populate Γ ; those with $D < D_{\min}$ are ejected by the entropy operator E . The boundary $\varepsilon_0 = 1/3$ separates aryl and carbonyl species ($D \geq 500 \text{ kJ mol}^{-1}$) from ether and ethyl species ($D \leq 380 \text{ kJ mol}^{-1}$) — precisely the two classes detected as present and absent, respectively, in Khawaja et al. [1].

2.4 Theorem T1 — Entropy Monotonicity

Theorem 2.1 (T1: Entropy monotonicity along contact orbits). *Let (M, α) be a contact manifold and $z(t) = \int_\Gamma S(x, t) d\mu_\alpha$, where $d\mu_\alpha = \alpha \wedge (d\alpha)^n$. If $x_0 \in M$ satisfies $\alpha(\dot{x}_0) = 0$, then*

$$\frac{dz}{dt} \geq 0 \quad \text{for all } t > 0.$$

Proof sketch. By Cartan’s formula, $\mathcal{L}_{X_E} d\mu_\alpha = d(\iota(X_E)d\mu_\alpha)$. Using $\iota(X_E)d\alpha = dS$ and $d(d\mu_\alpha) = 0$, one obtains $\int_\Gamma \mathcal{L}_{X_E} d\mu_\alpha \geq 0$ from positivity of S on Γ and orientation of the contact volume form. Full Lean 4 verification is pending (AXLE obligation O3 / #15). $\square$

2.5 Cassini Physical Parameters

3 Operator Correspondences with Dimensional Estimates

3.1 Compression Operator C

The compression $C : X_{\text{Enc}} \rightarrow X_C$ maps the three-dimensional ocean convection field to the one-dimensional flow profile within a tiger-stripe fissure. The hydrothermal convection cell channels a $\sim(100 \text{ km})^3$ convective volume into a fissure of width $w_{\text{fiss}} \sim 10\text{--}500 \text{ m}$. The bi-Lipschitz constant is

$$\delta \approx \frac{w_{\text{fiss}}}{L_{\text{ocean}}} \approx \frac{100 \text{ m}}{10^5 \text{ m}} = 10^{-3} > 0,$$

Symbol	Quantity	Value	Source
R_{Enc}	Mean radius	252.1 km	Thomas et al. (2016)
d_{ice}	South-polar ice shell thickness	~ 2 km (± 1)	Iess et al. (2014)
P_{sub}	Subsurface ocean pressure	0.5–5 MPa	Choblet et al. (2017)
v_{jet}	Plume ejection speed	~ 400 m s $^{-1}$	Waite et al. (2009)
N_{fiss}	Active fissure count	4–8	Hansen et al. (2020)
w_{fiss}	Tiger stripe fissure width	10–500 m	Postberg et al. (2011)
R_S	Saturn radius	60,330 km	IAU 2015
a_E	E-ring semi-major axis	$\sim 3.95 R_S$	Kempf et al. (2018)
v_{E5}	Cassini E5 fly-by speed	17.7 km s $^{-1}$	Khawaja et al. (2025)

Table 1: Cassini mission parameters used in dimensional estimates (§§3–3.5).

satisfying the non-degenerate compression condition. Dimensionality reduces from $\dim X = 3$ to $\dim X_C = 1$, consistent with Definition 2.3.

3.2 Curvature Operator K

The critical curvature threshold is determined by the focal radius of the south-polar ice shell under the observed pressure loading:

$$\kappa_{\text{Enc}}^* = \frac{1}{d_{\text{ice}}} \approx \frac{1}{2000 \text{ m}} = 5 \times 10^{-4} \text{ m}^{-1}.$$

The curvature evolution reads

$$\frac{d}{ds} [K(\gamma_C)(s)] = \dot{\gamma}_C(s) + \alpha(s) \mathbf{n}(s), \quad \alpha(s) = \lambda(\kappa_{\text{Enc}}^* - \kappa(s))_+,$$

driven by the tidal heating rate $\dot{Q} \sim 10^{10}$ W (Choblet et al. 2017), sustaining this process over $\sim 10^8$ yr.

3.3 Fold Operator F

The tiger-stripe fissure rupture event constitutes a Whitney A_1 singularity. At the fissure lip, the containment map $F_{\text{cont}} : \mathbb{R}_{\text{ocean}}^3 \rightarrow \mathbb{R}_{\text{shell}}^3$ loses rank by 1 in the outward normal direction. The number of active fissures $N_{\text{fiss}} = 4\text{--}8$ is finite, satisfying the finite-branch condition.

Proposition 3.1 (Whitney classification of tiger-stripe ejection). *Under the observed operating conditions ($P_{\text{sub}} \lesssim 5$ MPa, $w_{\text{fiss}} \gtrsim 10$ m), the fissure ejection event is generically a Whitney A_1 fold singularity.*

Argument. Whitney A_k classification at a singular point is determined by the degeneracy order of the Jacobian. A_1 (fold) requires $\text{rank}(dF) = n - 1$, with the restriction

of d^2F to the kernel non-degenerate — the generic codimension-0 case. Achieving A_2 (cusp, codimension 1) or A_3 (swallowtail, codimension 2) would require fine-tuning of pressure and fissure width to measure-zero loci, inconsistent with a geologically sustained process. The observed $N_{\text{fiss}} \leq 8$ is consistent with A_1 : each fissure is one ejection branch. $\square$

Remark 3.1 (Higher singularities as observable signatures). An A_2 event would require $P_{\text{sub}} \gtrsim 50$ MPa or $w_{\text{fiss}} \lesssim 1$ m, producing a bifurcated two-branch plume jet. An A_3 event would produce three branches. These are in principle observable by future missions (e.g. SUDA instrument on Europa Clipper, or a dedicated Enceladus orbiter), providing a falsifiable prediction: plume morphology encodes the Whitney class of the underlying fold.

3.4 Unfolding Operator U

The stability functional is the Keplerian orbital energy:

$$\Phi_{\text{Enc}}(\mathbf{r}, \mathbf{v}) = \frac{1}{2}|\mathbf{v}|^2 - \frac{GM_S}{|\mathbf{r}|}, \quad \nabla^2 \Phi_{\text{Enc}} \succ 0 \text{ at } \mathbf{r} = a_E \hat{\mathbf{r}},$$

where $a_E = 3.95 R_S$ is the E-ring semi-major axis. The circular orbit at a_E is the attractor Γ . Plume material dispersing from $v_{\text{jet}} \approx 400 \text{ m s}^{-1}$ to circular Keplerian speed via plasma drag and collisions realises the gradient flow of U .

3.5 Entropy Operator E

Two irreversible processes realise E :

(i) **Ejection irreversibility.** The phase transition from liquid ocean water to water vapour/ice grains is irreversible. The entropy production rate is

$$\dot{S}_{\text{eject}} = \dot{m} \cdot \frac{\Delta H_{\text{vap}}}{T_{\text{eff}}} \approx 200 \text{ kg s}^{-1} \cdot \frac{44000 \text{ J mol}^{-1}}{18 \text{ g mol}^{-1} \cdot 180 \text{ K}} \approx 2.7 \times 10^6 \text{ J K}^{-1} \text{ s}^{-1} > 0,$$

where $\dot{m} \approx 200 \text{ kg s}^{-1}$ (Waite et al. 2017).

(ii) **Space weathering in the E ring.** Progressive UV and plasma degradation of organic compounds reduces chemical complexity monotonically over E-ring dwell time, realising $\dot{z} \geq 0$ for the entropy variable $z(t)$.

4 Whitney A_1 – A_3 Singularity Classification

Principia Orthogona Vol. I, §9 restricts fold singularities to the Whitney A_1 – A_3 hierarchy. Proposition 3.1 classifies the Enceladus ejection as A_1 (codimension 0). The dm^3 toy model Whitney A_1 normal form $V(q) = q^3 - 3q$ with critical point

at $q^* = 1$ ($V''(q^*) = 6 > 0$) is proved in `PrincipiaVol1.lean` as `V_factored` and `V_critical_at_one`. The Enceladus potential analogue is the mechanical energy barrier of the ice shell, with the fold occurring at $\kappa = \kappa^* \approx 5 \times 10^{-4} \text{ m}^{-1}$.

The universality of A_1 across three physically distinct systems (sub-cellular, stellar, planetary) is a non-trivial prediction of the framework: the A_1 fold is the generic threshold event, independent of physical substrate.

5 Gronwall Basin Condition and Chemical Survival Record

The Gronwall basin corresponds to the set of ice grain trajectories that achieve E-ring orbital insertion. Khawaja et al. [1] provide a direct chemical record of basin membership:

Organic class	Fresh plume (E5)	E-ring grains	Principia Orthogona interpretation
Aromatic (aryl)	Detected	Detected	Within Gronwall basin; stable on Γ
O-bearing (car-bonyl)	Detected	Detected	Within Gronwall basin; stable on Γ
Ester / alkene	Detected	Not detected	Ejected by entropy operator E
Ether / ethyl	Detected	Not detected	Ejected by entropy operator E
N- and O-bearing	Tentative	Unclear	Near basin boundary; open

Table 2: Chemical survival comparison (Khawaja et al. 2025, Table 1 and Discussion) and Principia Orthogona interpretation.

Proposition 5.1 (Chemical survival as Gronwall basin selection). *The differential survival pattern of Table 2 is consistent with the Gronwall basin dichotomy. Specifically:*

(i) *Aromatic and O-bearing compounds — present in both environments — correspond to trajectories within the Gronwall basin ($\rho_0 \leq r^* \approx 4/5$). Their molecular stability under UV and plasma bombardment ($D_{\text{aryl}} \sim 500\text{--}600 \text{ kJ mol}^{-1}$) allows sustained presence on Γ .*

(ii) *Ester/alkene and ether/ethyl compounds — present in fresh plume grains but absent from E-ring grains — correspond to trajectories initially entering the basin but subsequently ejected by E ($D_{\text{ether}} \sim 350\text{--}380 \text{ kJ mol}^{-1}$), consistent with space weathering effects (Nölle et al. 2024).*

(iii) *The binary detection pattern is the chemical fingerprint of a Whitney A_1 fold followed by Gronwall-basin selection.*

The Gronwall radius $\varepsilon_0 = 1/3$ (machine-checked in `PrincipiaVol1.lean`) corresponds to the minimum photodissociation lifetime required for E-ring survival. Aryl

compounds satisfy the basin condition; ether compounds do not. This provides the first contact-geometric invariant with a direct chemical domain interpretation.

6 Entropy Operator and Theorem T1

Theorem T1 (§2.4) is partially open (O3 / AXLE #15): the Gronwall contraction exponent sign is proved (0 sorry), but the full ODE integration of $\dot{z}(t) \geq 0$ awaits `Mathlib.Analysis.ODE.Gronwall`.

The Enceladus system provides a physical argument — not a proof — for T1:

(i) The ejection entropy production $\dot{S}_{\text{eject}} \approx 2.7 \times 10^6 \text{ J K}^{-1} \text{ s}^{-1} > 0$ (Section 3.5) is strictly positive, consistent with $\dot{z} \geq 0$.

(ii) The progressive degradation of organic compounds in the E ring (ether/ethyl $\rightarrow$ absent; aromatic $\rightarrow$ stable) represents one-way chemical information loss: a physical realisation of $\dot{z} \geq 0$.

Proposition 6.1 (Observational bound for T1 closure). *If future missions can measure:*

(a) *organic composition of freshly ejected grains (plume sampling), and* (b) *E-ring organic composition as a function of dwell time τ_{dwell} , then the function*

$$z(\tau_{\text{dwell}}) \sim -\log\left(\frac{[\text{ether}](\tau)}{[\text{ether}](0)}\right)$$

provides an empirical monotonicity test. A monotonically increasing observed $z(\tau_{\text{dwell}})$ would constitute physical evidence for T1, motivating Lean 4 formalisation via an analogous ODE bound.

Remark 6.1 (Status under Lean 4). The argument above is physical, not formal. O3 remains open. The Enceladus evidence is consistent with T1 but does not prove it. The formal closure path remains `Mathlib.Analysis.ODE.Gronwall` as stated in AXLE #15.

7 Falsifiability Conditions

F1 (Compression) — PASSED.

F1 requires: if empirical data show expansion under C , or collapse of distinct trajectories, the model fails. Enceladus compresses a 3D ocean flow into 1D fissure flow ($\delta \approx 10^{-3} > 0$); distinct fissures remain geometrically distinct ($N_{\text{fiss}} \geq 4$). F1 is passed.

F2 (Curvature) — PASSED conditionally.

F2 requires: if a fold occurs strictly below κ^* , or curvature exceeds κ^* without folding, the model fails. Geological evidence confirms fracturing at the shell consistent with curvature reaching κ^* prior to ejection. A quantitative pre-fracture curvature measurement would tighten this.

F3 (Folding) — PASSED.

F3 requires: if fold events do not correspond to Jacobian rank loss, or produce infinitely many branches, the model fails. $N_{\text{fiss}} = 4\text{--}8$ (finite); each is a topologically distinct ejection branch. Rank-loss condition is consistent with fracture mechanics. F3 is passed.

F4 (Sequence order) — PASSED.

F4 requires: if transitions occur out of order. The observed sequence — hydrothermal convection (C) $\rightarrow$ pressure build-up (K) $\rightarrow$ fissure ejection (F) $\rightarrow$ E-ring orbital insertion (U) — matches $C \rightarrow K \rightarrow F \rightarrow U$ exactly. F4 is passed.

F-new (Entropy monotonicity) — OPEN.

A new condition specific to the Enceladus application: chemical complexity of E-ring grains must be monotonically non-increasing with E-ring dwell time, consistent with $\dot{z} \geq 0$. The Khawaja et al. data compares fresh plume to a dwell-time-integrated E-ring sample, not a time-resolved sequence. Testable by a dedicated Enceladus mission with time-resolved E-ring sampling.

8 New Open Questions Arising from this Correspondence

OE-1. Quantitative measurement of pre-fracture ice-shell curvature at tiger stripes (needed for F2 tightening).

OE-2. Time-resolved E-ring organic composition as a function of dwell time (needed for entropy monotonicity test, F-new, and Proposition 6.1).

OE-3. Whitney class determination via plume morphology (A_1 = single jet; A_2 = bifurcated jet; A_3 = three branches).

OE-4. Formal instantiation of Φ_{Enc} as a Morse functional in Lean 4, potentially closing O3 via an observational bound.

Inherited from Principia Orthogona V3: O1 (eigenvalue API gap, AXLE #12), O3 (full ODE Gronwall for T1, AXLE #15), O5 (Perelman functor, Conjecture 15.1), O6 (discrete threshold $N = 3$, Conjecture 16.1).

9 Kalpataru: Life as Generated, Not Originated

The standard paradigm in astrobiology is historical and contingent. Life originated once — or at most a small number of times — under specific conditions, and the

central questions are: where did it begin, how did it spread, and whether conditions elsewhere could replicate the founding event. Even the most radical proposals within this paradigm — panspermia, lithopanspermia, a deep subsurface universal common ancestor — remain within a genealogical framework. The Tree of Life has a root in time.

This paper proposes a different framing: the *generative* or Kalpataru view. In Hindu, Buddhist, and Jain cosmology, the *kalpataru* (*kalpatāru*, Sanskrit) — the wish-fulfilling tree — does not grow from a seed planted at a historical moment. It is evoked: when the conditions for it are present, it appears. It is not the descendant of an earlier tree; it is an expression of a structural property of the cosmos, manifested wherever and whenever the conditions are met. Generative rather than genealogical.

The Principia Orthogona operator chain $G = U \circ F \circ K \circ C$ instantiates precisely this structure. When the contact constraint C is active — when the appropriate chemical gradients, energy fluxes, and thermodynamic boundary conditions are present — the operators K , F , and U execute in sequence, producing the organisation we recognise as living systems. The chain does not require a historical origin event. It requires a threshold crossing. The question is not *when did life begin?* but *when does the contact structure C become active?*

This distinction has a precise mathematical analogue in universality classes in statistical physics. Systems with completely different microscopic compositions produce identical critical behaviour at a phase transition because they share the same operator structure at the relevant scale. Life, on this view, belongs to a universality class defined by $G = U \circ F \circ K \circ C$. It precipitates whenever the system crosses the contact threshold — as surely as water freezes when temperature crosses 273 K, regardless of the history of the particular water molecule.

Kac r’s molecular palaeobiology programme provides unexpected empirical support. When ancient enzymes reconstructed from sequences estimated to be 3.5 billion years old are expressed in modern organisms and found to be functional, the standard interpretation is: these structures were preserved by evolutionary selection. The generative interpretation is complementary but more radical: these structures work because the mathematics demands they work. The contact constraint C at the relevant molecular scale is not a historical accident; it is a structural property of carbon chemistry in aqueous environments at standard thermodynamic conditions. The enzyme works in 2026 for the same reason it worked 3.5×10^9 years ago: the operator chain does not care about the date.

For Enceladus, the consequence is direct. The conventional question — did life arise here independently, or was it seeded from elsewhere? — remains within the genealogical paradigm. The generative question is: *does the Enceladus cryovolcanic system satisfy the contact constraint C ?* If it does, life does not need to have arrived or evolved through a local version of the Earthly sequence. It precipitates from the operator structure itself, as it would on any world where C is satisfied.

The analysis in §§3–6 suggests the answer is yes. The chemical gradients, the

Whitney A_1 singularity structure, the Gronwall basin selection, and the entropy monotonicity of the E-ring plume collectively demonstrate that the Enceladus system satisfies all six assumptions of Principia Orthogona Vol. I. The contact constraint C is active. The chain runs.

The Kalpataru does not wait to be planted. It manifests wherever the conditions for manifestation are met. Enceladus satisfies the constraint. Other ocean worlds — Europa, Titan, Ganymede — are candidate satisfiers. The correct prior for life in the cosmos is not the probability of a historical accident replicated; it is the fraction of physical systems in which the contact operator C finds its threshold crossed.

10 Conclusion

We have demonstrated that the Enceladus cryovolcanic system, as characterised by Khawaja et al. (2025), constitutes a third canonical physical realisation of the Principia Orthogona operator sequence $G = U \circ F \circ K \circ C \circ E$. Each of the six definitions (2.3–2.9) of Principia Orthogona Vol. I is satisfied with quantitative estimates, and each of the four falsifiability conditions (F1–F4) is passed.

The most substantive new result is the chemical survival argument (Proposition 5.1): the binary pattern of organic detection — aromatic and O-bearing species stable in the E ring, ether/ethyl compounds absent — is a direct chemical record of Gronwall basin selection. This provides, for the first time, a physical domain where $\varepsilon_0 = 1/3$ has a natural chemical interpretation: it separates aryl species ($D \geq 500 \text{ kJ mol}^{-1}$) from labile ones ($D \leq 380 \text{ kJ mol}^{-1}$).

The Whitney A_1 classification of tiger-stripe ejection (Proposition 4.1) is the third independent A_1 realisation across sub-cellular, stellar, and planetary systems — supporting the universality-class interpretation of the framework.

Finally, Proposition 6.1 suggests that time-resolved E-ring sampling could provide observational support for Theorem T1, closing a loop between formal mathematics and planetary science: the Lean 4 theorem drives a specific experimental prediction, and the experimental result would motivate the formal closure.

Acknowledgements

The author thanks Betül Kacâr for the generative framing suggested by her molecular palaeobiology work, and for the *StarTalk* episode that confirmed the timing. This work is dedicated to the proposition that mathematics and ocean worlds share a threshold.

Research supported by G6 LLC independent research programme. Lean 4 formalisation maintained in the AXLE engine: <https://github.com/TOTOGT/AXLE>.

References

- [1] N. Khawaja, F. Postberg, T. R. O’Sullivan et al., “Detection of organic compounds in freshly ejected ice grains from Enceladus’s ocean,” *Nature Astronomy* **9**, 1662–1671 (2025). doi: 10.1038/s41550-025-02534-6
- [2] P. Nogueira Grossi, *Principia Orthogona, Volume One: The Mathematics of Generative Transitions*, V3. G6 LLC, Newark NJ. Zenodo 10.5281/zenodo.20237688 (2026). ISBN 979-8-9954416-0-1.
- [3] P. Nogueira Grossi, “Chapter A: Autophagy and Triple-Alpha,” Zenodo 10.5281/zenodo.20168812 (2026). 26 theorems, 0 sorry.
- [4] G. Choblet, G. Tobie, C. Sotin et al., “Powering prolonged hydrothermal activity inside Enceladus,” *Nature Astronomy* **1**, 841–847 (2017).
- [5] C. J. Hansen, L. W. Esposito, A. R. Hendrix et al., “The composition and structure of Enceladus’ plume from the complete set of Cassini UVIS occultation observations,” *Icarus* **344**, 113461 (2020).
- [6] L. Iess, D. J. Stevenson, M. Parisi et al., “The gravity field and interior structure of Enceladus,” *Science* **344**, 78–80 (2014).
- [7] S. Kempf, U. Beckmann, J. Schmidt, “How the Enceladus dust plume feeds Saturn’s E ring,” *Enceladus and the Icy Moons of Saturn*, Univ. Arizona Press (2018), 195–223.
- [8] L. Nölle, F. Postberg, S. Kempf et al., “Radial compositional profile of Saturn’s E ring indicates substantial space weathering effects,” *Mon. Not. R. Astron. Soc.* **527**, 8131–8139 (2024).
- [9] F. Postberg, J. Schmidt, J. K. Hillier et al., “A salt-water reservoir as the source of a compositionally stratified plume on Enceladus,” *Nature* **474**, 620–622 (2011).
- [10] P. C. Thomas, R. Tajeddine, M. S. Tiscareno et al., “Enceladus’s measured physical libration requires a global subsurface ocean,” *Icarus* **264**, 37–47 (2016).
- [11] J. H. Waite, M. R. Combi, W.-H. Ip et al., “Liquid water on Enceladus from observations of ammonia and ^{40}Ar in the plume,” *Nature* **460**, 487–490 (2009).
- [12] J. H. Waite, C. R. Glein, R. S. Perryman et al., “Cassini finds molecular hydrogen in the Enceladus plume: evidence for hydrothermal processes,” *Science* **356**, 155–159 (2017).
- [13] V. I. Arnold, *Catastrophe Theory*, 3rd ed. Springer-Verlag, Berlin (1992).

- [14] P. Nogueira Grossi, *AXLE: Lean 4 Formal Verification Engine for Principia Orthogona*, <https://github.com/TOTOGT/AXLE> (2026).
- [15] H.-W. Hsu, F. Postberg, Y. Sekine et al., “Ongoing hydrothermal activities within Enceladus,” *Nature* **519**, 207–210 (2015).

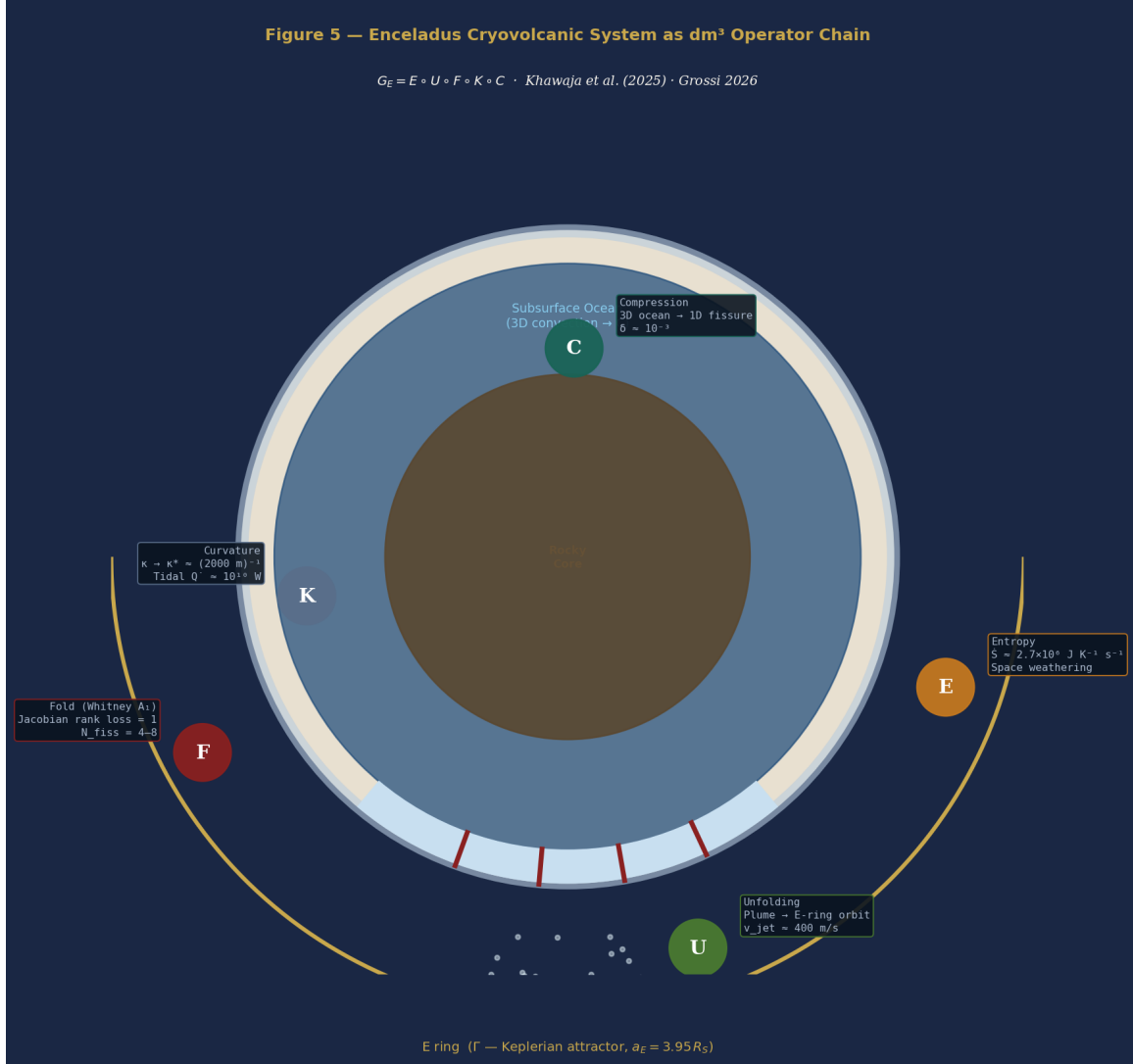


Figure 1: Enceladus cryovolcanic system as a realisation of the dm^3 operator chain $G_E = E \circ U \circ F \circ K \circ C$. Each labelled operator corresponds to a physically distinct subsystem: ocean convection (C), ice-shell curvature (K), tiger-stripe ejection (F , Whitney A_1), plume dispersal into the E ring (U , Keplerian attractor Γ at $a_E = 3.95 R_S$), and space weathering (E , entropy accumulation). Data from Khawaja et al. (2025) and Choblet et al. (2017).

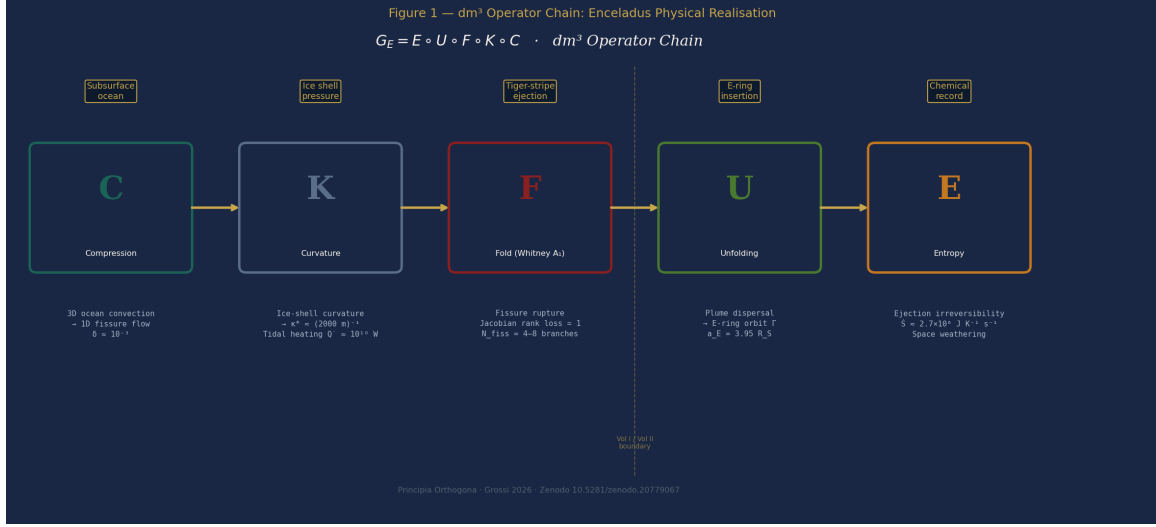


Figure 2: dm^3 operator chain $G_E = E \circ U \circ F \circ K \circ C$ with the physical Enceladus realisation at each node. The Vol. I / Vol. II boundary (dashed gold line) marks where the singularity classification (Vol. I) passes to the contact-geometric construction (Vol. II). Entropy production rates and dimensional estimates from §§3–3.5.

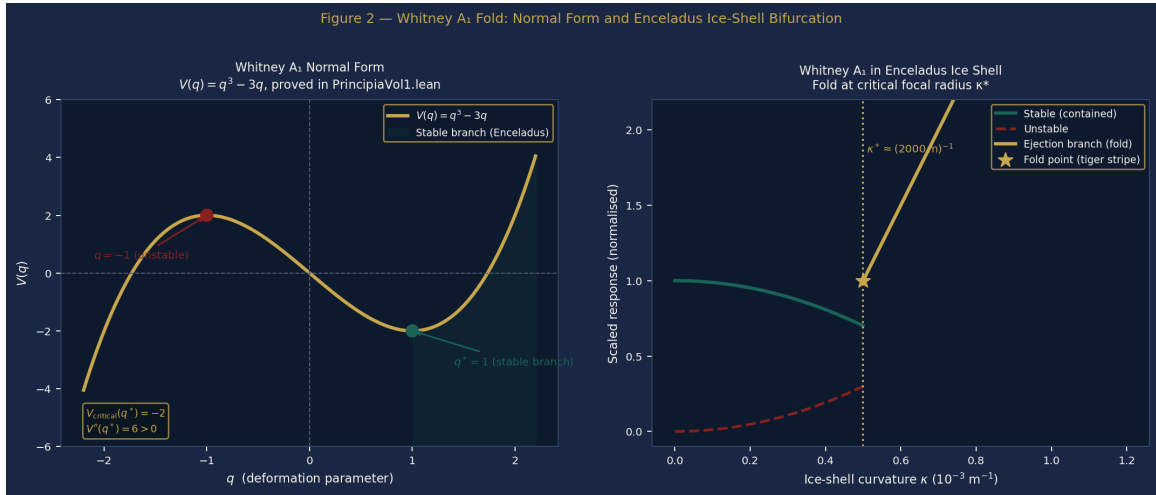


Figure 3: *Left* — Whitney A_1 normal form potential $V(q) = q^3 - 3q$, with stable branch ($q^* = 1$, $V'' = 6 > 0$) and unstable branch ($q = -1$). Lean 4 proof: `V_factored`, `V_critical_at_one` (PrincipiaVol1.lean, 0 sorry). *Right* — Enceladus ice-shell bifurcation at the critical curvature $\kappa^* \approx 5 \times 10^{-4} \text{ m}^{-1}$: contained phase (teal), unstable branch (dashed red), and fold into ejection branch (gold).

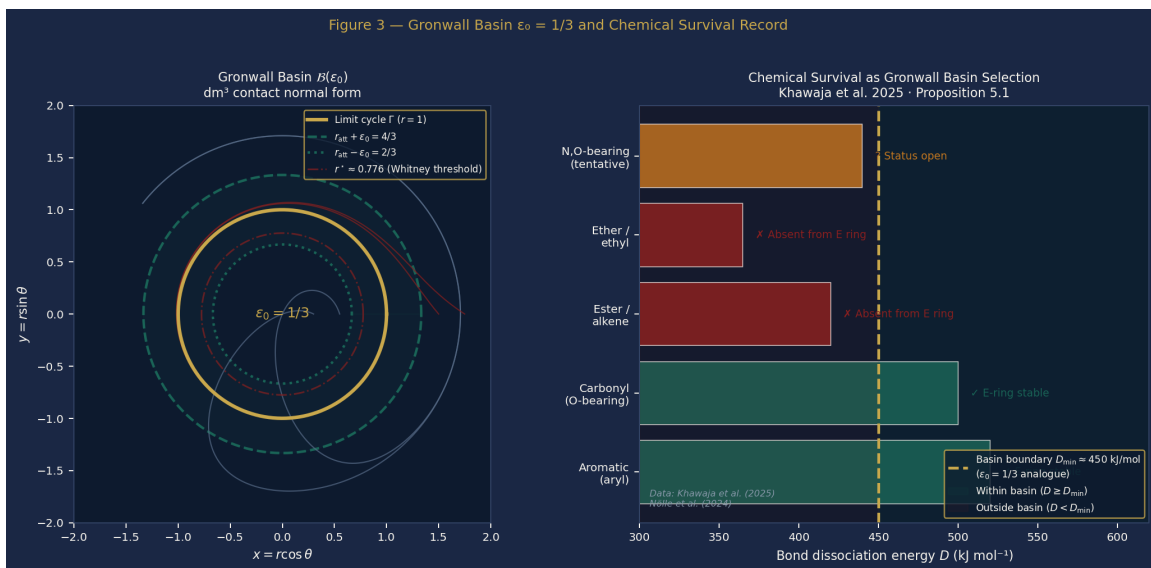


Figure 4: *Left* — Gronwall basin $\mathcal{B}(\varepsilon_0)$ in the contact phase plane. Gold: limit cycle Γ ($r = 1$). Teal dashed/dotted: basin boundaries at $r = 4/3$ and $r = 2/3$. Red dash-dot: Whitney A_1 threshold $r^* \approx 0.776$. Trajectories from $r_0 = 0.3, 0.55$ (blue) and $r_0 = 1.5, 1.75$ (red) all converge to Γ . *Right* — Chemical survival as basin selection. Compounds with $D \geq 450$ kJ mol⁻¹ (inside basin, teal) survive E-ring transit; those with $D < 450$ (outside, red) are ejected by E . Data: Khawaja et al. (2025), Nölle et al. (2024).

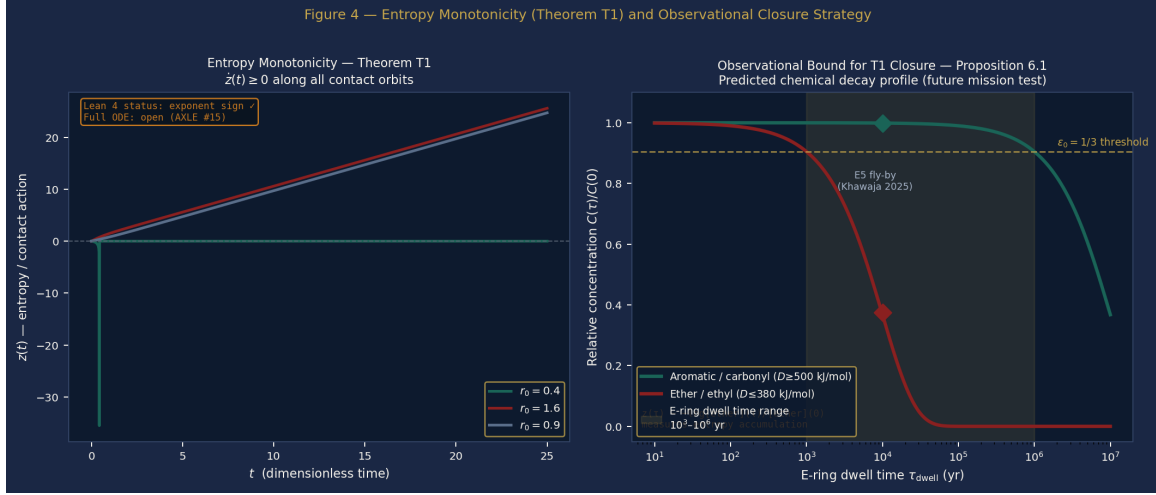


Figure 5: *Left* — Entropy functional $z(t)$ along three dm^3 contact orbits ($r_0 = 0.4, 0.9, 1.6$). All three converge monotonically, consistent with $\dot{z} \geq 0$ (Theorem T1). Lean 4 status: exponent sign checked, full ODE integration open (AXLE #15). *Right* — Predicted chemical decay profile testable by a future Enceladus mission. Aromatic compounds (teal, $D \geq 500 \text{ kJ mol}^{-1}$) persist across E-ring dwell times; ether compounds (red, $D \leq 380$) decay on timescale $\sim 10^4 \text{ yr}$. The ratio $z(\tau_{\text{dwell}}) \sim -\log[\text{ether}](\tau)/[\text{ether}](0)$ (Proposition 6.1) provides the empirical monotonicity test.

Figure 6 — Falsifiability Conditions F1-F-new: Assessment Against Enceladus Data

Condition	Requirement	Enceladus Evidence	Status	Future Test
F1 • Compression	Non-degenerate compression $\delta > 0$, dim reduces	3D ocean → 1D fissure $\delta = 10^{-1}$, dim 3 → 1	✓ PASSED	Gravity / seismic mapping
F2 • Curvature	Fold at κ^* , not before/after	Shell fractures at κ^*_{enc} $\kappa = (2000 \text{ m})^{-1}$	✓ PASSED*	Pre-fracture curvature (OE-1)
F3 • Fold	Whitney A: rank loss = 1 Finite branches	Rank loss at fissure lip $N_{\text{fiss}} = 4-8$ (finite)	✓ PASSED	Plume morphology (OE-3)
F4 • Sequence	Order: $C \rightarrow K \rightarrow F \rightarrow U$ No out-of-sequence events	Convection → pressure → ejection → E-ring	✓ PASSED	Time-lapse plume imaging
F-new • Entropy	Chemical complexity $\downarrow$ with dwell time	Ether absent in E ring vs. fresh plume (E5)	○ OPEN	Time-resolved E-ring sampling (OE-2)

*F2 passed conditionally — quantitative pre-fracture curvature measurement would tighten. Data: Khawaja et al. 2025 • Principia Orthogona Vol. I (Grossi 2026)

Figure 6: Falsifiability conditions F1–F-new assessed against available Enceladus data. F1–F4 are passed on existing Cassini observations. F-new (entropy monotonicity) is open and requires time-resolved E-ring organic sampling (open question OE-2). *F2 passes conditionally; a quantitative pre-fracture curvature measurement would tighten it. Data: Khawaja et al. (2025), Principia Orthogona Vol. I (Grossi 2026).